

W00 · Setup · 30 MINIT / MINUTES · Hari 1, 09:00 hingga 09:30

Hands-On SQA-AI Assistant Workshop

Set up your SQA-AI assistant

Sediakan pembantu AI SQA anda

Clone the course repository, check the PC, install opencode, start it in the lab folder with the free model and learn the five controls you will use all day.

opencode 1.18 · model Big Pickle (OpenCode Zen, free, no API key) · synthetic ShopFast data only

Bengkel SQA KRISAv2 dan PPrISA 2.0 · 14 hingga 15 September 2026 · Makmal Komputer Aras 11, Menara MyIPO

[Web guide \(copy prompts\)](#)

[Module hub](#)

[Example outputs \(GitHub\)](#)

Before you start / Sebelum mula

YOU WILL / ANDA AKAN

1. Step 0: install Git and Node.js if they are missing
2. Clone the repository and run setup\check-lab.ps1
3. Install opencode with npm and check the version
4. Start opencode in lab\ with the Big Pickle model
5. Learn Tab (Plan or Build), @file, /command, /models and !command

1. Langkah 0: pasang Git dan Node.js jika belum ada
2. Klon repositori dan jalankan setup\check-lab.ps1
3. Pasang opencode dengan npm dan semak versi
4. Mulakan opencode dalam lab\ dengan model Big Pickle
5. Kenali Tab (Plan atau Build), @fail, /arahan, /models dan !arahan

CHECK FIRST / SEMAK DAHULU

- PowerShell is in C:\Users\<you>\sqa-krisa-bengkel\lab
- opencode starts and shows Big Pickle, OpenCode Zen (W00)

Data rules / Peraturan data

- Synthetic ShopFast data only. Never paste real agency documents or personal data.
- Free models run outside Malaysia and may learn from prompts.
- AI output is a draft. A person checks and signs off.

OUTPUT

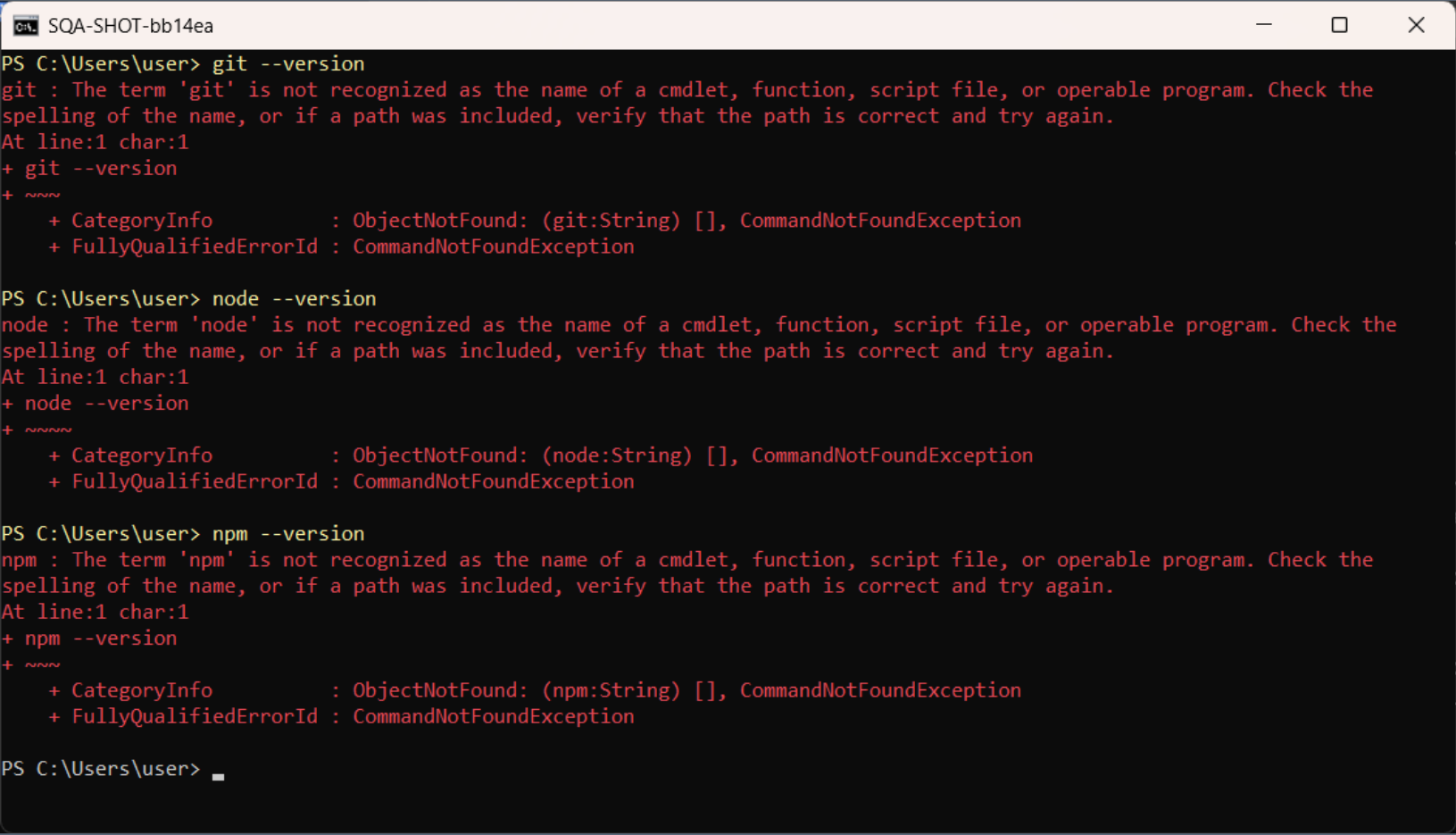
PC sedia: check-lab.ps1 menunjukkan SEMUA OK

Open PowerShell (Start, type PowerShell, press Enter). Check whether Git and Node.js are installed. If all three show a version (node v20 or newer), skip to Step 1.

Buka PowerShell (Start, taip PowerShell, tekan Enter). Semak sama ada Git dan Node.js sudah dipasang. Jika ketiga-tiga memaparkan versi (node v20 atau lebih baharu), terus ke Langkah 1.

TYPE IN POWERSHELL

```
git --version
node --version
npm --version
```



```
SQA-SHOT-bb14ea
PS C:\Users\user> git --version
git : The term 'git' is not recognized as the name of a cmdlet, function, script file, or operable program. Check the
spelling of the name, or if a path was included, verify that the path is correct and try again.
At line:1 char:1
+ git --version
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (git:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

PS C:\Users\user> node --version
node : The term 'node' is not recognized as the name of a cmdlet, function, script file, or operable program. Check the
spelling of the name, or if a path was included, verify that the path is correct and try again.
At line:1 char:1
+ node --version
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (node:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

PS C:\Users\user> npm --version
npm : The term 'npm' is not recognized as the name of a cmdlet, function, script file, or operable program. Check the
spelling of the name, or if a path was included, verify that the path is correct and try again.
At line:1 char:1
+ npm --version
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (npm:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

PS C:\Users\user> _
```

Seeing 'is not recognized'? On a PC with admin rights, install both with winget. Click Yes if Windows asks for permission. Then close PowerShell and open it again.

Nampak 'is not recognized'? Pada PC dengan hak admin, pasang kedua-duanya dengan winget. Klik Yes jika Windows meminta kebenaran. Kemudian tutup PowerShell dan buka semula.

TYPE IN POWERSHELL

```
winget install -e --id Git.Git --accept-source-
agreements --accept-package-agreements
winget install -e --id OpenJS.NodeJS.LTS --accept-
source-agreements --accept-package-agreements
```

EXPECTED OUTPUT / JANGKAAN OUTPUT (version numbers may differ)

```
Found Git [Git.Git] Version 2.55.x
This application is licensed to you by its owner.
Downloading https://github.com/git-for-windows/...
Successfully verified installer hash
Starting package install...
Successfully installed
```

```
Found Node.js (LTS) [OpenJS.NodeJS.LTS] Version 24.x
...
Successfully installed
```

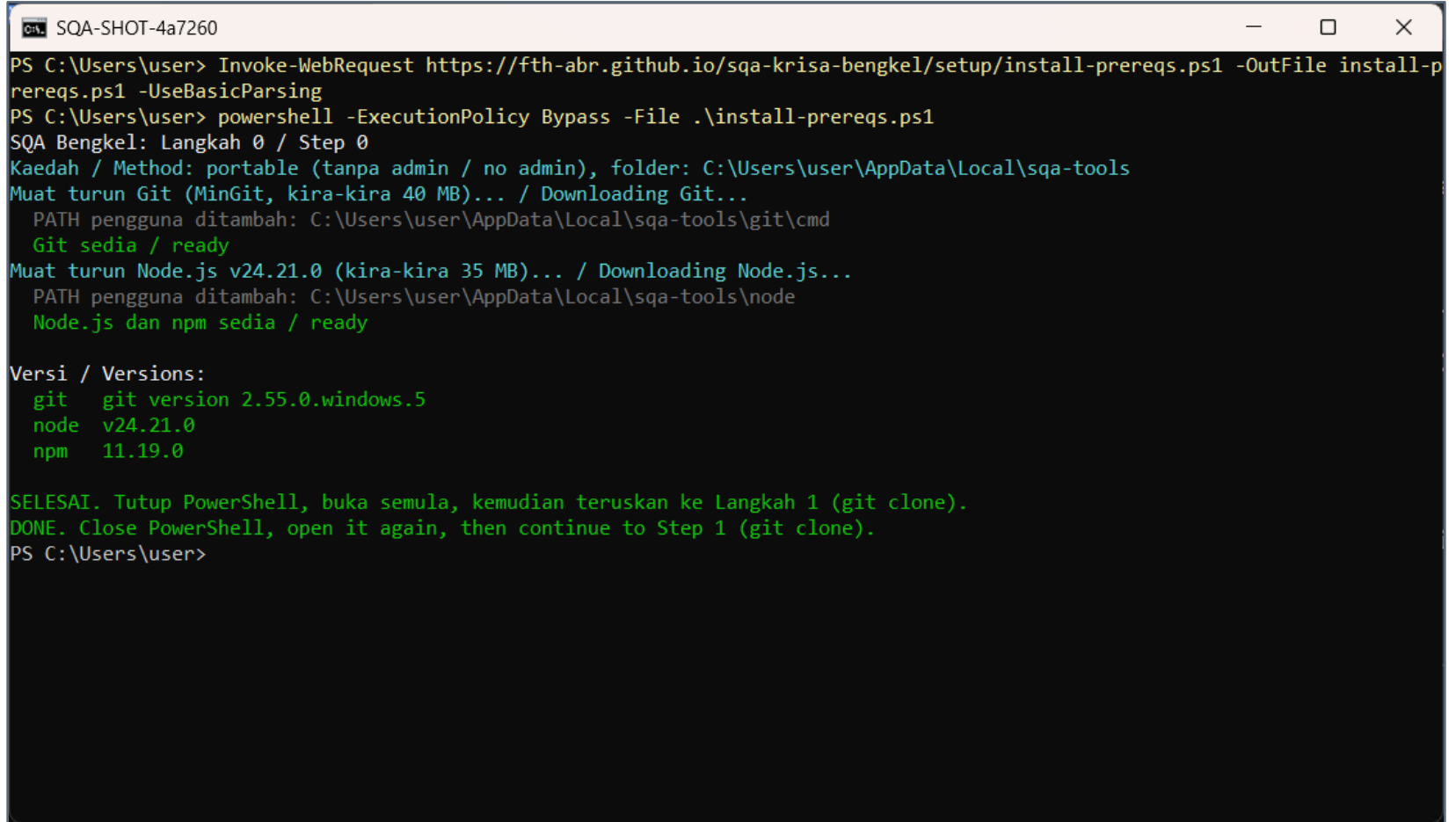
After both installs: close PowerShell, open a new window, then do Step 0D.
Selepas kedua-duanya dipasang: tutup PowerShell, buka tetingkap baharu, kemudian buat Langkah 0D.
No admin rights or winget missing? Go to Step 0C.

No winget, no admin rights, or winget failed? Download the course script and run it. It installs portable Git and Node.js in your own user folder. No admin rights needed.

Tiada winget, tiada hak admin, atau winget gagal? Muat turun skrip kursus dan jalankan. Ia memasang Git dan Node.js versi portable dalam folder pengguna anda. Hak admin tidak diperlukan.

TYPE IN POWERSHELL

```
cd $HOME
Invoke-WebRequest https://fth-abr.github.io/sqa-krisa-bengkel/setup/install-prereqs.ps1 -OutFile install-prereqs.ps1 -UseBasicParsing
powershell -ExecutionPolicy Bypass -File .\install-prereqs.ps1
```



```
SQA-SHOT-4a7260
PS C:\Users\user> Invoke-WebRequest https://fth-abr.github.io/sqa-krisa-bengkel/setup/install-prereqs.ps1 -OutFile install-prereqs.ps1 -UseBasicParsing
PS C:\Users\user> powershell -ExecutionPolicy Bypass -File .\install-prereqs.ps1
SQA Bengkel: Langkah 0 / Step 0
Kaedah / Method: portable (tanpa admin / no admin), folder: C:\Users\user\AppData\Local\sqa-tools
Muat turun Git (MinGit, kira-kira 40 MB)... / Downloading Git...
  PATH pengguna ditambah: C:\Users\user\AppData\Local\sqa-tools\git\cmd
  Git sedia / ready
Muat turun Node.js v24.21.0 (kira-kira 35 MB)... / Downloading Node.js...
  PATH pengguna ditambah: C:\Users\user\AppData\Local\sqa-tools\node
  Node.js dan npm sedia / ready

Versi / Versions:
git    git version 2.55.0.windows.5
node   v24.21.0
npm    11.19.0

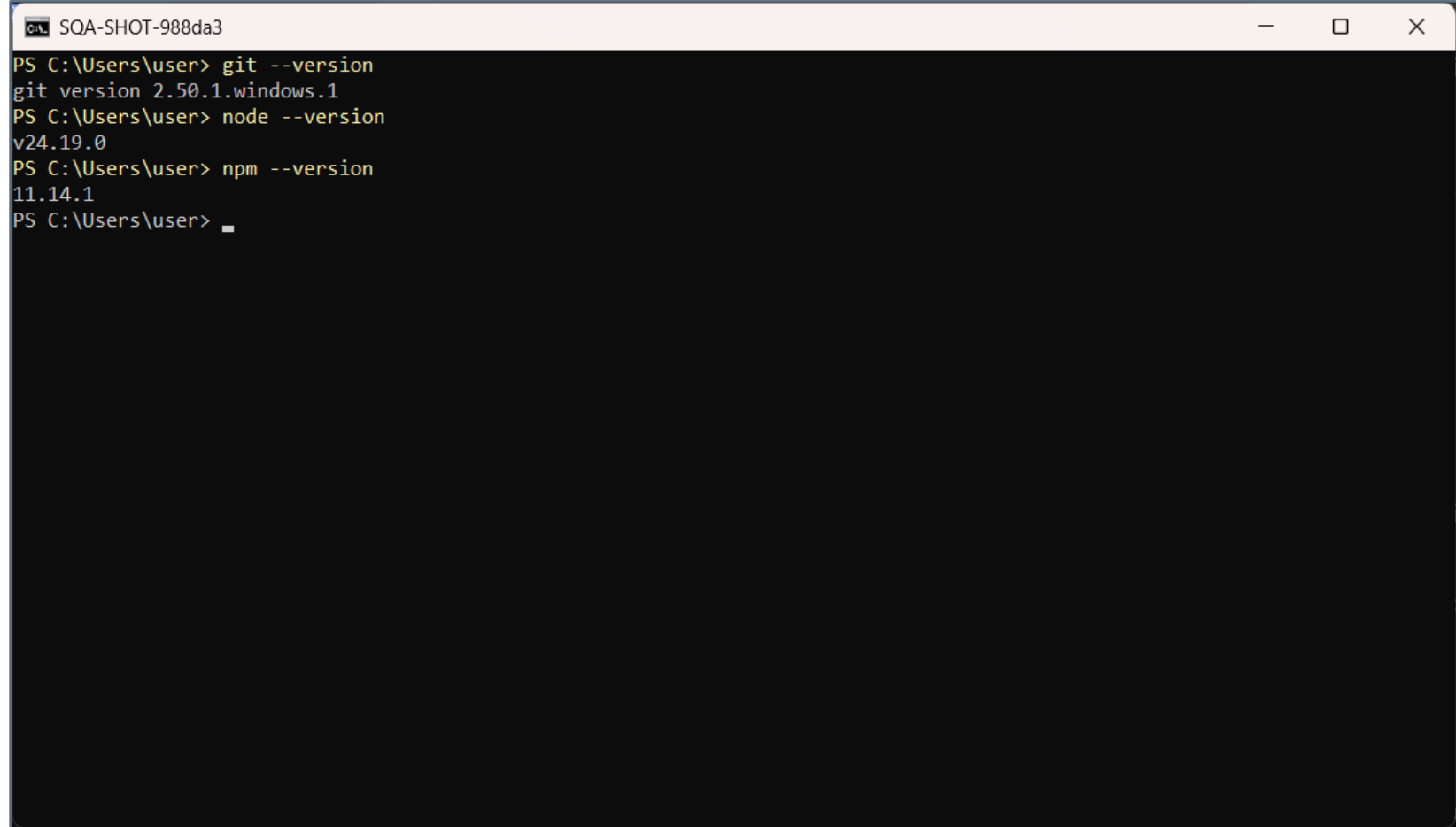
SELESAI. Tutup PowerShell, buka semula, kemudian teruskan ke Langkah 1 (git clone).
DONE. Close PowerShell, open it again, then continue to Step 1 (git clone).
PS C:\Users\user>
```

Close PowerShell and open a new window so PATH refreshes. Check again. All three must show a version before you continue to Step 1.

Tutup PowerShell dan buka tetingkap baharu supaya PATH dikemas kini. Semak semula. Ketiga-tiga mesti memaparkan versi sebelum meneruskan ke Langkah 1.

TYPE IN POWERSHELL (NEW WINDOW)

```
git --version
node --version
npm --version
```



```
PS C:\Users\user> git --version
git version 2.50.1.windows.1
PS C:\Users\user> node --version
v24.19.0
PS C:\Users\user> npm --version
11.14.1
PS C:\Users\user> █
```

Step 0 problems / Masalah Langkah 0

git or node is still not recognised

Close ALL PowerShell windows and open a new one. Still failing: run `install-prereqs.ps1` again (Step 0C)

winget is not recognised

Skip 0B. Use Step 0C: `install-prereqs.ps1` needs no winget and no admin rights

Invoke-WebRequest fails (TLS or network)

`[Net.ServicePointManager]::SecurityProtocol = 'Tls12'` then run it again. Behind a proxy, ask the trainer for the proxy address

Windows asks for admin and you have none

Click No, then use Step 0C (portable, no admin)

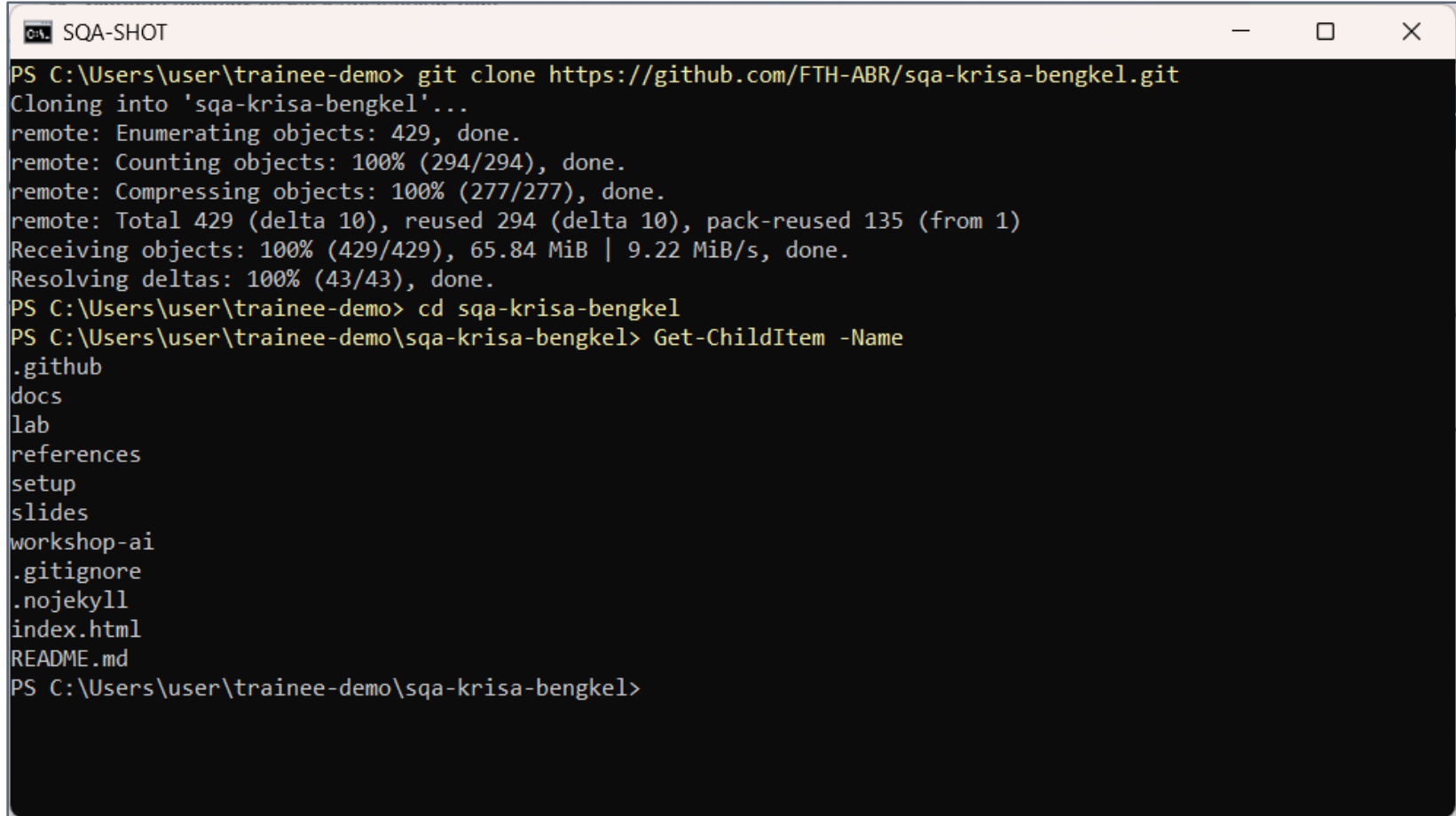
When Step 0D shows three versions, continue to Step 1 (git clone).

Open PowerShell, go to your home folder and clone the course repository.

Buka PowerShell, pergi ke folder rumah dan klon repositori kursus.

TYPE IN POWERSHELL

```
cd $HOME
git clone https://github.com/FTH-ABR/sqa-krisa-bengkel.git
cd sqa-krisa-bengkel
```



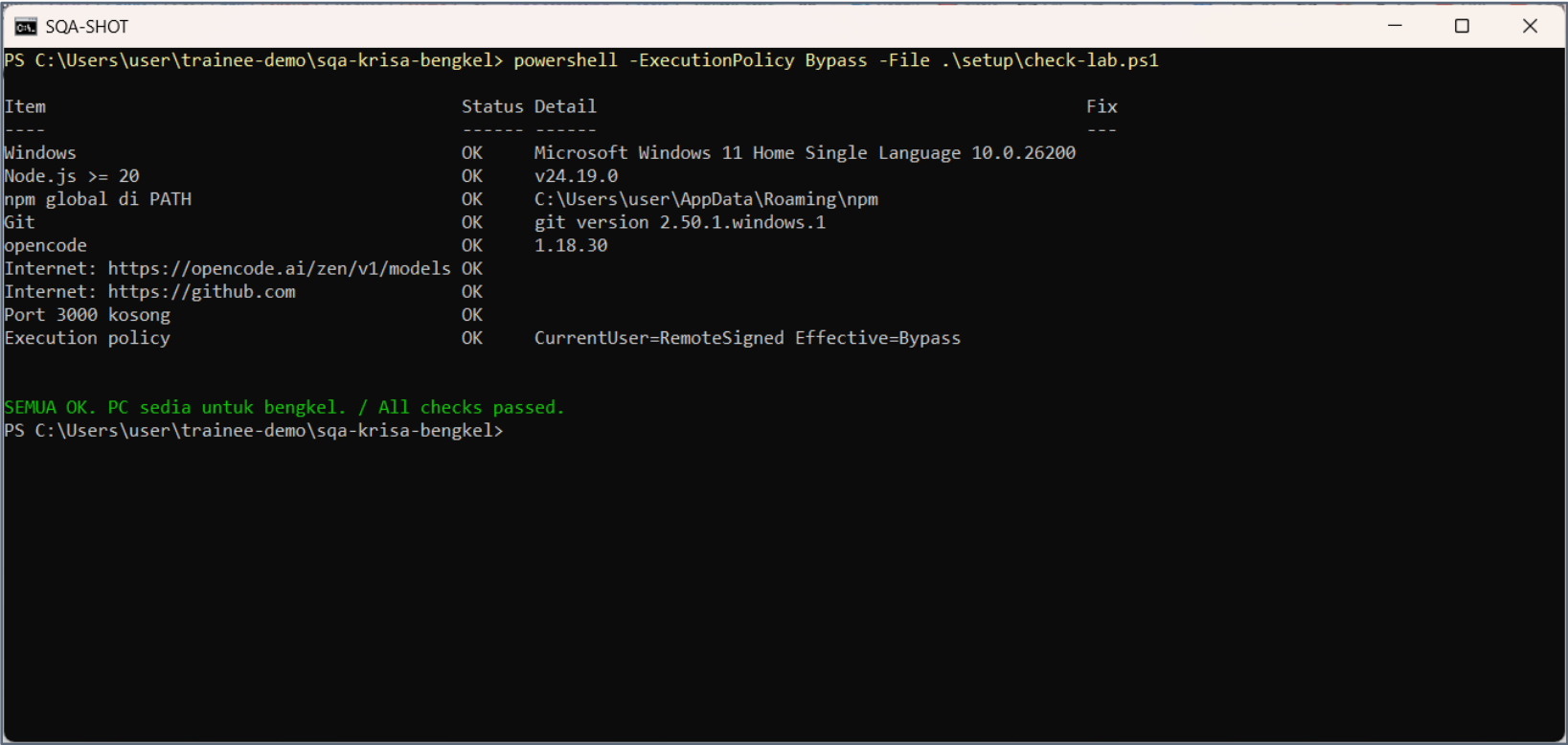
```
SQA-SHOT
PS C:\Users\user\trainee-demo> git clone https://github.com/FTH-ABR/sqa-krisa-bengkel.git
Cloning into 'sqa-krisa-bengkel'...
remote: Enumerating objects: 429, done.
remote: Counting objects: 100% (294/294), done.
remote: Compressing objects: 100% (277/277), done.
remote: Total 429 (delta 10), reused 294 (delta 10), pack-reused 135 (from 1)
Receiving objects: 100% (429/429), 65.84 MiB | 9.22 MiB/s, done.
Resolving deltas: 100% (43/43), done.
PS C:\Users\user\trainee-demo> cd sqa-krisa-bengkel
PS C:\Users\user\trainee-demo\sqa-krisa-bengkel> Get-ChildItem -Name
.github
docs
lab
references
setup
slides
workshop-ai
.gitignore
.nojekyll
index.html
README.md
PS C:\Users\user\trainee-demo\sqa-krisa-bengkel>
```

Run the lab readiness check. Every line must say OK. Fix anything else using the Fix column.

Jalankan semakan kesediaan makmal. Setiap baris mesti OK. Betulkan yang lain menggunakan lajur Fix.

TYPE IN POWERSHELL

```
powershell -ExecutionPolicy Bypass -File
.\setup\check-lab.ps1
```

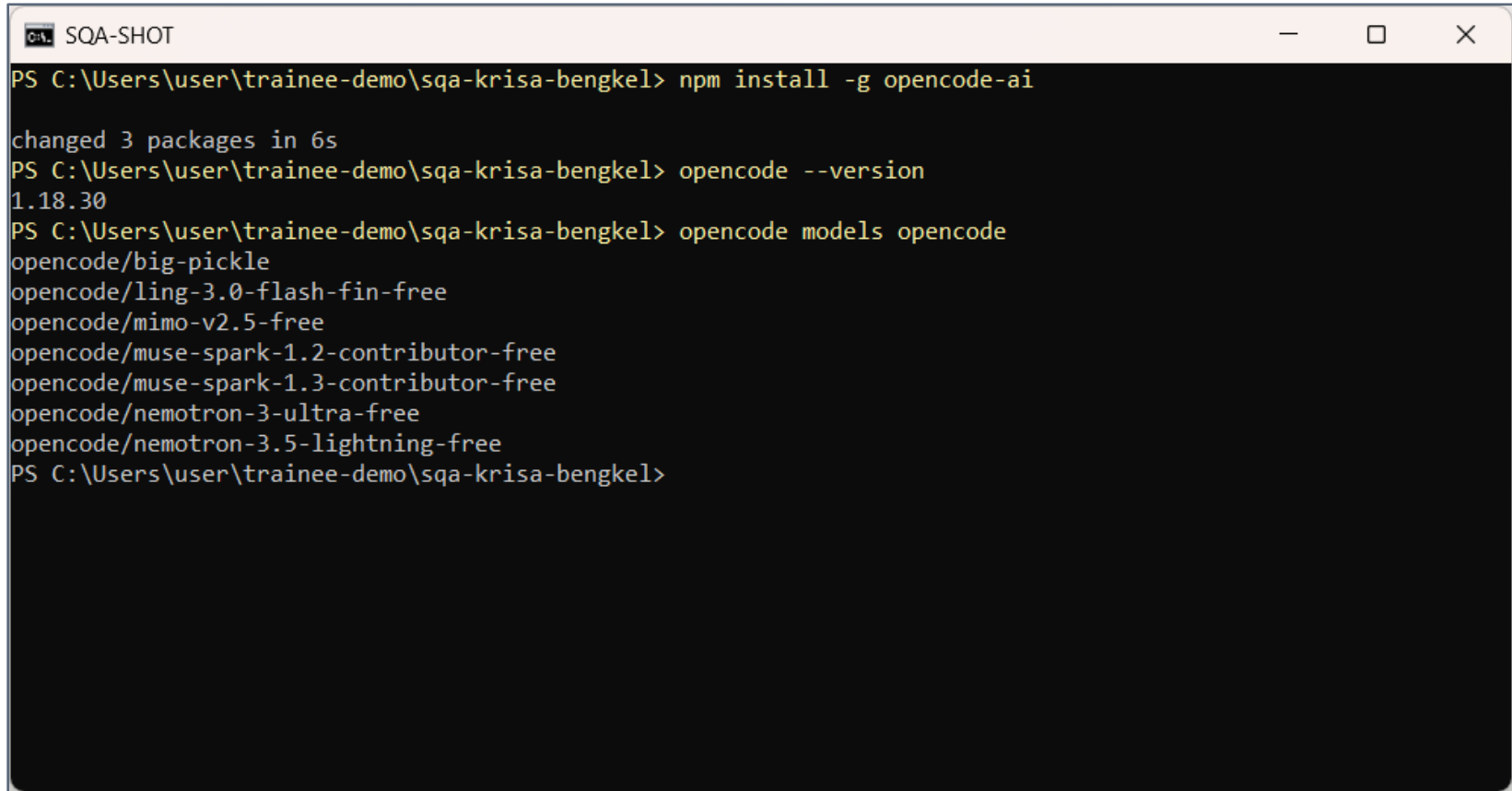


Install opencode with npm, confirm the version and list the free OpenCode Zen models.

Pasang opencode dengan npm, sahkan versi dan senaraikan model percuma OpenCode Zen.

TYPE IN POWERSHELL

```
npm install -g opencode-ai
opencode --version
opencode models opencode
```



```
PS C:\Users\user\trainee-demo\sqa-krisa-bengkel> npm install -g opencode-ai

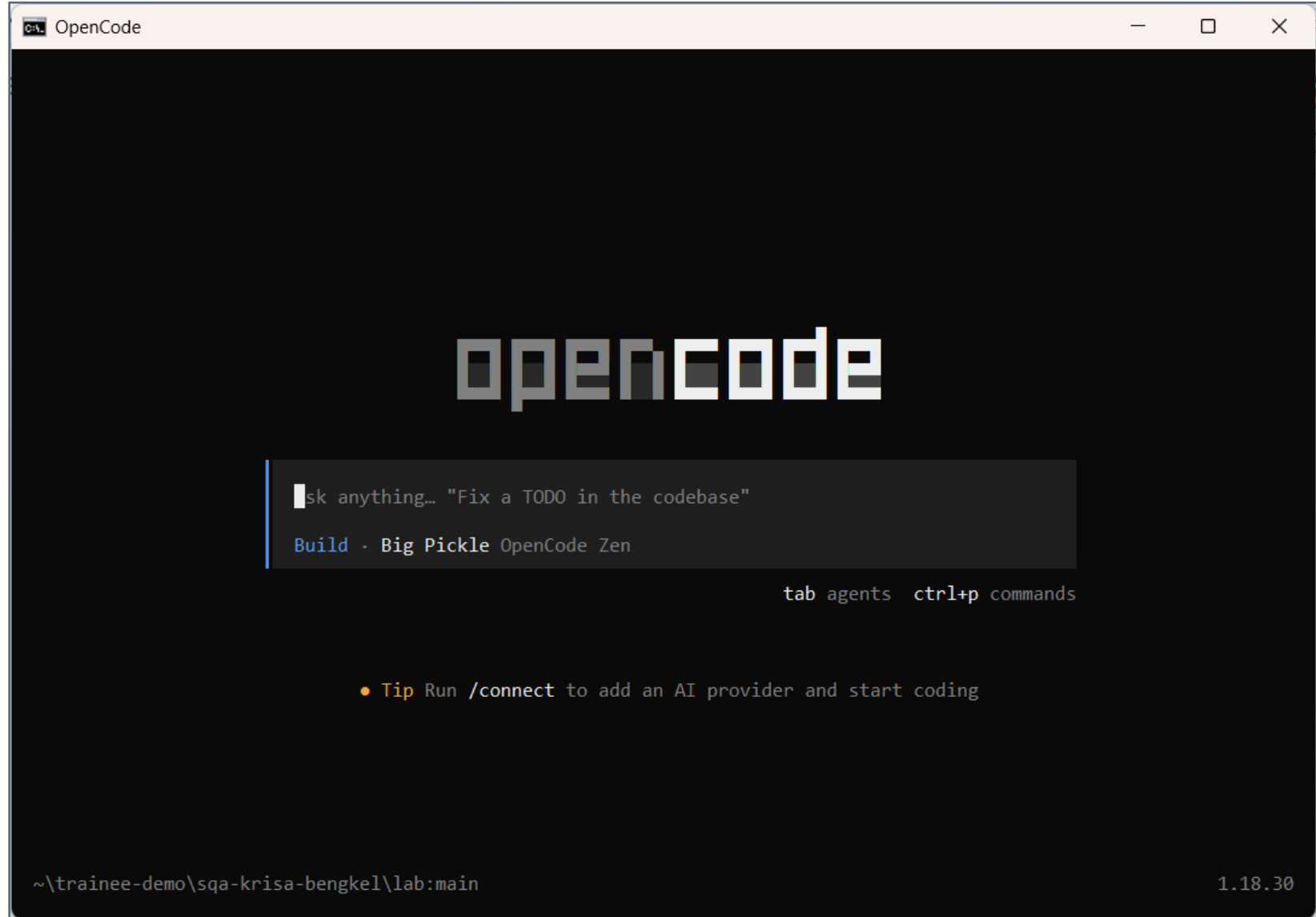
changed 3 packages in 6s
PS C:\Users\user\trainee-demo\sqa-krisa-bengkel> opencode --version
1.18.30
PS C:\Users\user\trainee-demo\sqa-krisa-bengkel> opencode models opencode
opencode/big-pickle
opencode/ling-3.0-flash-fin-free
opencode/mimo-v2.5-free
opencode/muse-spark-1.2-contributor-free
opencode/muse-spark-1.3-contributor-free
opencode/nemotron-3-ultra-free
opencode/nemotron-3.5-lightning-free
PS C:\Users\user\trainee-demo\sqa-krisa-bengkel>
```

Go into the lab folder and start opencode with the free Big Pickle model. No account or API key is needed.

Masuk ke folder lab dan mulakan opencode dengan model percuma Big Pickle. Tiada akaun atau kunci API diperlukan.

TYPE IN POWERSHELL

```
cd lab  
opencode -m opencode/big-pickle
```

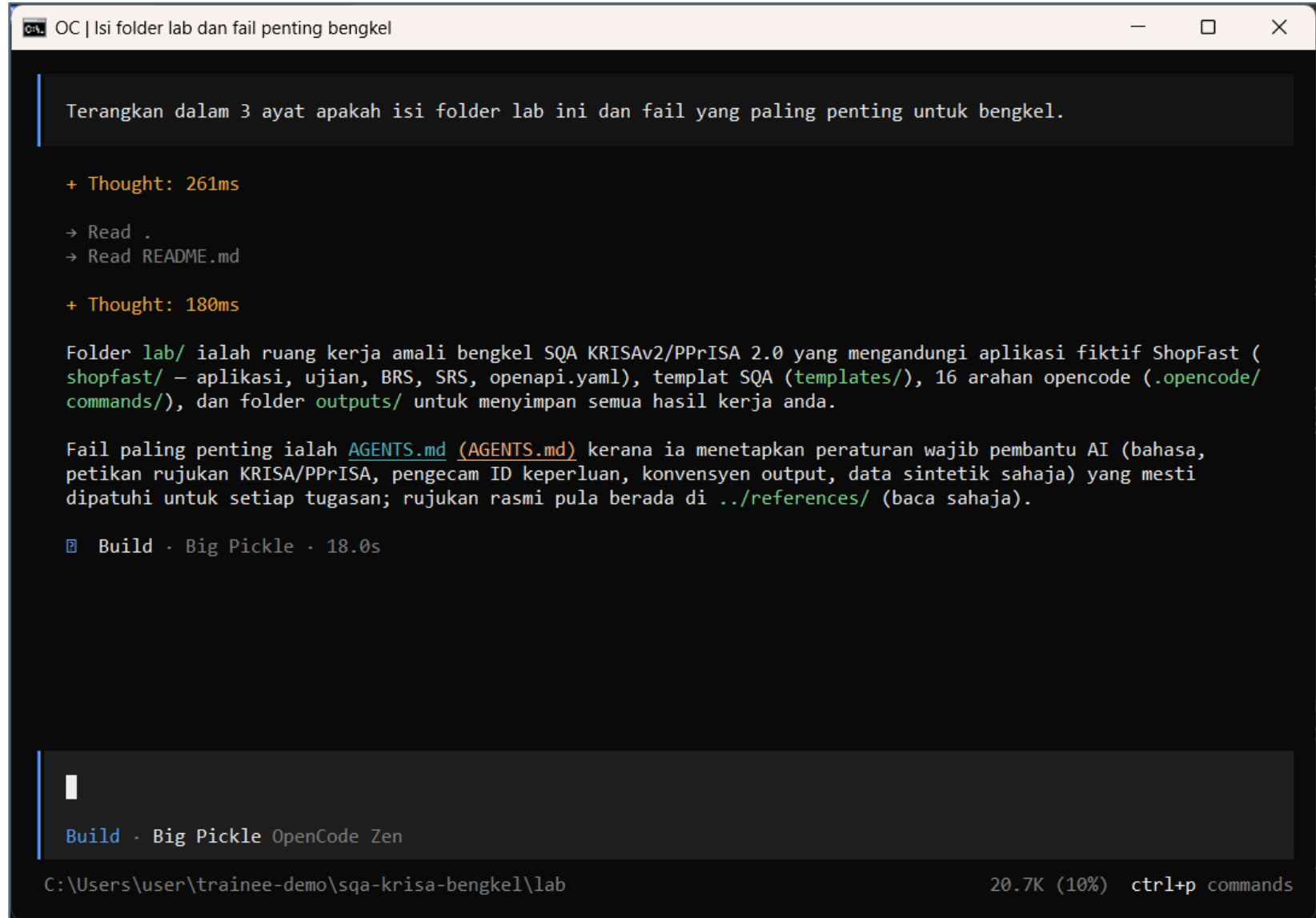


Type your first question and press Enter. The assistant reads the folder and AGENTS.md before answering.

Taip soalan pertama dan tekan Enter. Pembantu membaca folder dan AGENTS.md sebelum menjawab.

TYPE IN OPENCODE

Terangkan dalam 3 ayat apakah isi folder lab ini dan fail yang paling penting untuk bengkel.



The screenshot shows the OpenCode IDE window titled "OC | Isi folder lab dan fail penting bengkel". The chat interface displays a user query and the AI's response. The AI's response includes a thought process, a list of files read, another thought process, and a detailed explanation of the folder structure and the importance of the AGENTS.md file. The status bar at the bottom shows the file path, file size, and keyboard shortcuts.

```
OC | Isi folder lab dan fail penting bengkel

Terangkan dalam 3 ayat apakah isi folder lab ini dan fail yang paling penting untuk bengkel.

+ Thought: 261ms
→ Read .
→ Read README.md

+ Thought: 180ms

Folder lab/ ialah ruang kerja amali bengkel SQA KRISAv2/PPrISA 2.0 yang mengandungi aplikasi fiktif ShopFast (shopfast/ – aplikasi, ujian, BRS, SRS, openapi.yaml), templat SQA (templates/), 16 arahan opencode (.opencode/commands/), dan folder outputs/ untuk menyimpan semua hasil kerja anda.

Fail paling penting ialah AGENTS.md (AGENTS.md) kerana ia menetapkan peraturan wajib pembantu AI (bahasa, petikan rujukan KRISA/PPrISA, pengecam ID keperluan, konvensyen output, data sintetik sahaja) yang mesti dipatuhi untuk setiap tugas; rujukan rasmi pula berada di ../references/ (baca sahaja).

Build - Big Pickle - 18.0s

Build - Big Pickle OpenCode Zen

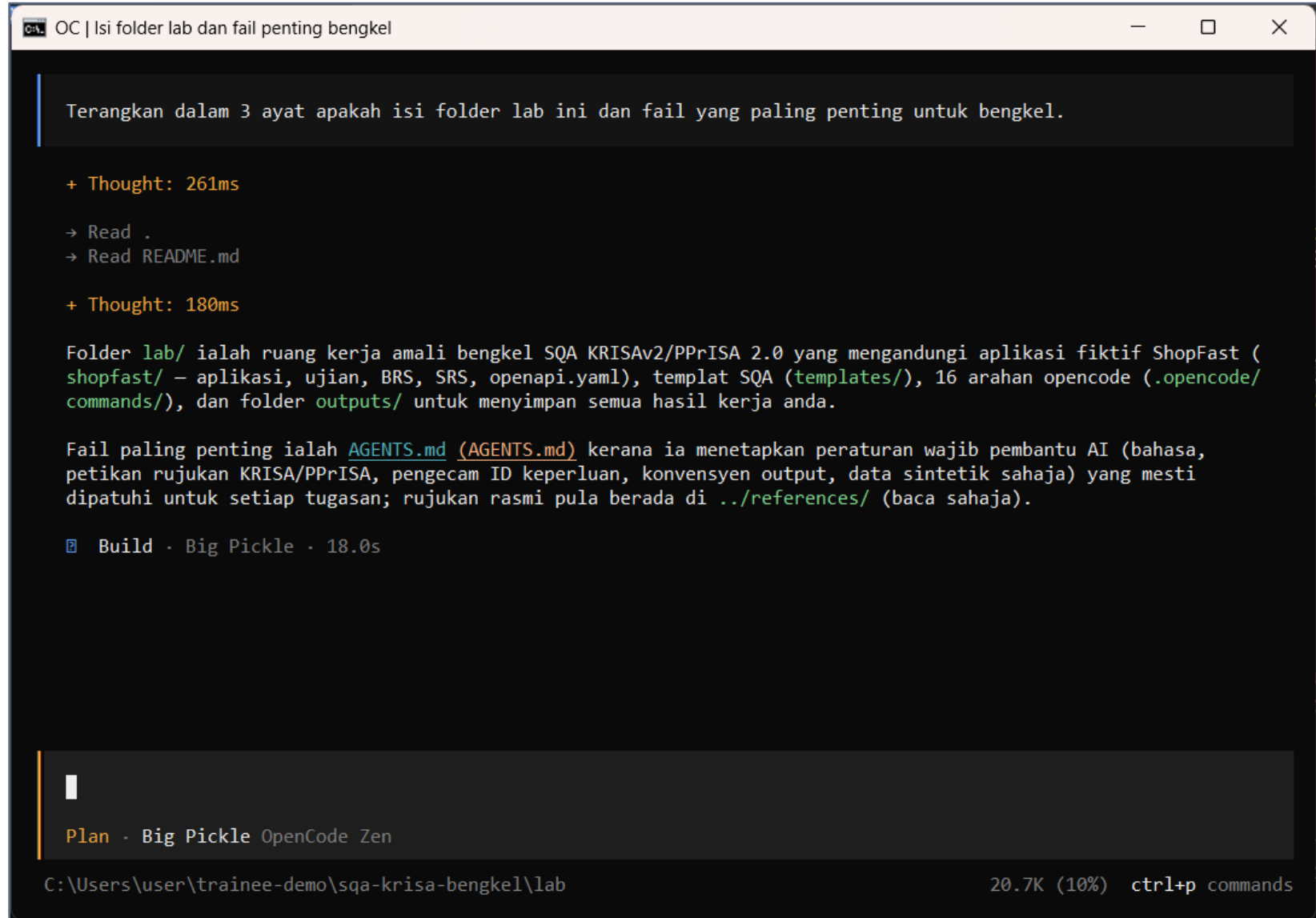
C:\Users\user\trainee-demo\sqa-krisa-bengkel\lab 20.7K (10%) ctrl+p commands
```

Press Tab to switch between Build (can edit files) and Plan (read only). Use Plan for reviews.

Tekan Tab untuk bertukar antara Build (boleh ubah fail) dan Plan (baca sahaja). Guna Plan untuk semakan.

TYPE IN OPENCODE

Tab



OC | Isi folder lab dan fail penting bengkel

Terangkan dalam 3 ayat apakah isi folder lab ini dan fail yang paling penting untuk bengkel.

+ Thought: 261ms

→ Read .
→ Read README.md

+ Thought: 180ms

Folder `lab/` ialah ruang kerja amali bengkel SQA KRISAv2/PPrISA 2.0 yang mengandungi aplikasi fiktif ShopFast (`shopfast/` – aplikasi, ujian, BRS, SRS, `openapi.yaml`), templat SQA (`templates/`), 16 arahan opencode (`.opencode/commands/`), dan folder `outputs/` untuk menyimpan semua hasil kerja anda.

Fail paling penting ialah `AGENTS.md` (`AGENTS.md`) kerana ia menetapkan peraturan wajib pembantu AI (bahasa, petikan rujukan KRISA/PPrISA, pengecam ID keperluan, konvensyen output, data sintetik sahaja) yang mesti dipatuhi untuk setiap tugas; rujukan rasmi pula berada di `../references/` (baca sahaja).

Build · Big Pickle · 18.0s

Plan · Big Pickle OpenCode Zen

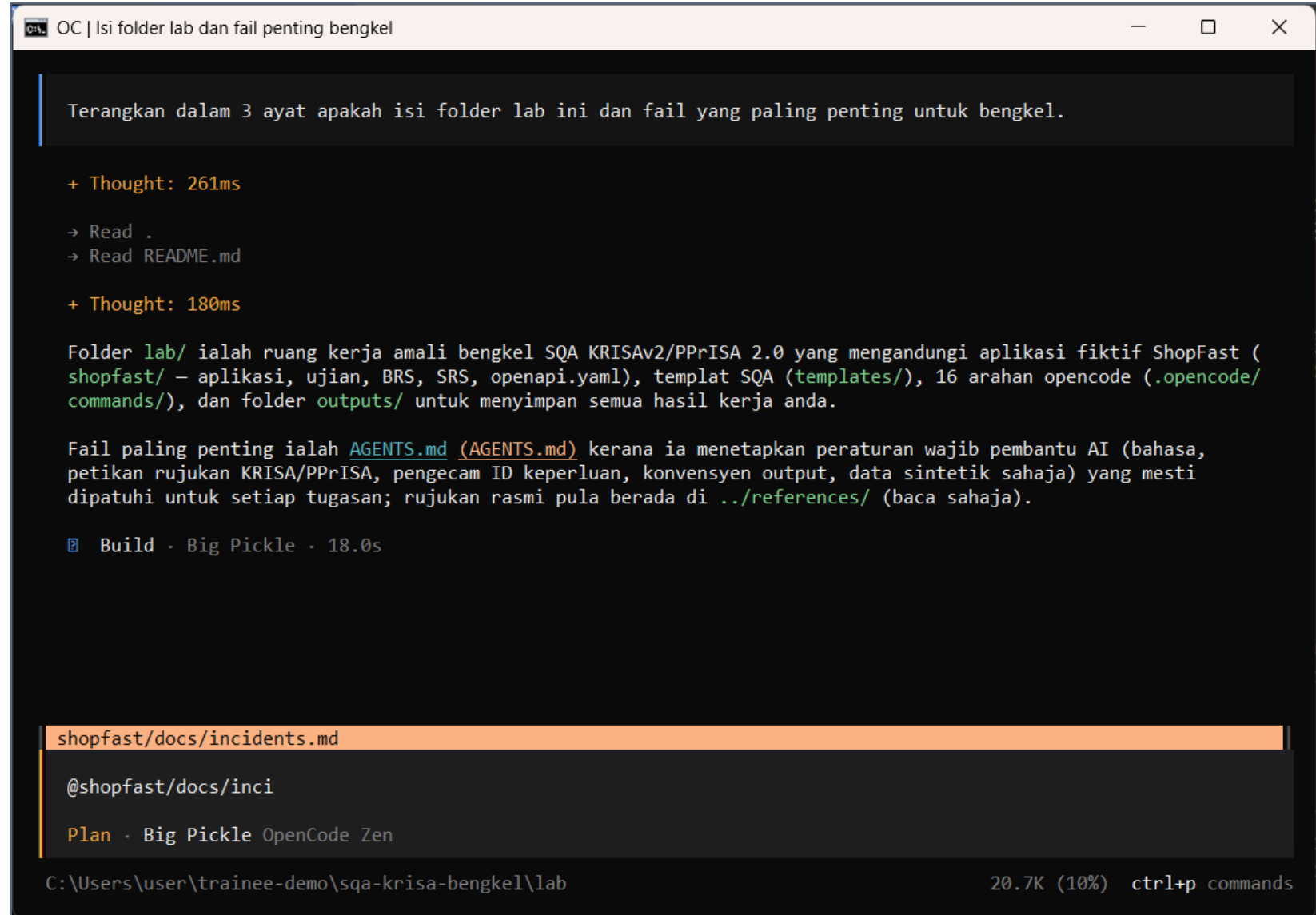
C:\Users\user\trainee-demo\sqa-krisa-bengkel\lab 20.7K (10%) `ctrl+p` commands

Type @ and part of a file name to attach a file to your prompt. Pick it from the list.

Taip @ dan sebahagian nama fail untuk melampirkan fail pada prompt. Pilih daripada senarai.

TYPE IN OPENCODE

```
@shopfast/docs/inci
```



The screenshot shows the OpenCode IDE window titled "OC | Isi folder lab dan fail penting bengkel". The chat window contains the following text:

Terangkan dalam 3 ayat apakah isi folder lab ini dan fail yang paling penting untuk bengkel.

+ Thought: 261ms

→ Read .
→ Read README.md

+ Thought: 180ms

Folder `lab/` ialah ruang kerja amali bengkel SQA KRISAv2/PPrISA 2.0 yang mengandungi aplikasi fiktif ShopFast (`shopfast/` – aplikasi, ujian, BRS, SRS, `openapi.yaml`), templat SQA (`templates/`), 16 arahan opencode (`.opencode/commands/`), dan folder `outputs/` untuk menyimpan semua hasil kerja anda.

Fail paling penting ialah `AGENTS.md` (`AGENTS.md`) kerana ia menetapkan peraturan wajib pembantu AI (bahasa, petikan rujukan KRISA/PPrISA, pengecam ID keperluan, konvensyen output, data sintetik sahaja) yang mesti dipatuhi untuk setiap tugas; rujukan rasmi pula berada di `../references/` (baca sahaja).

Build · Big Pickle · 18.0s

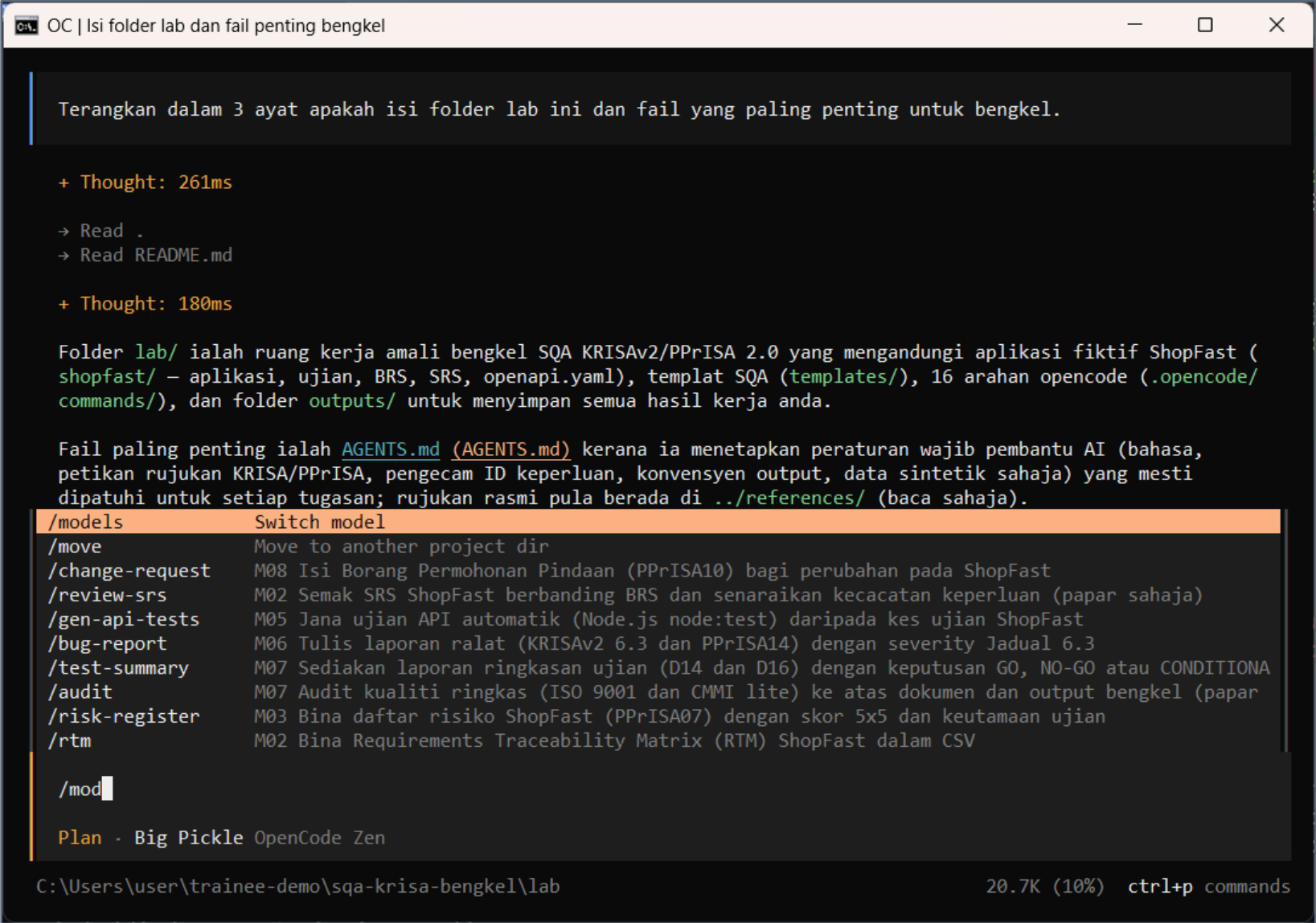
The file list at the bottom shows `shopfast/docs/incidents.md` selected, with `@shopfast/docs/inci` entered in the search bar. The status bar at the bottom indicates the current file is `C:\Users\user\trainee-demo\sqa-krisa-bengkel\lab`, with a size of 20.7K (10%) and a keyboard shortcut of `ctrl+p` for commands.

Type / to see commands. The course adds 16 SQA commands such as /review-srs and /testcases.

Taip / untuk melihat arahan. Kursus menambah 16 arahan SQA seperti /review-srs dan /testcases.

TYPE IN OPENCODE

```
/mod
```

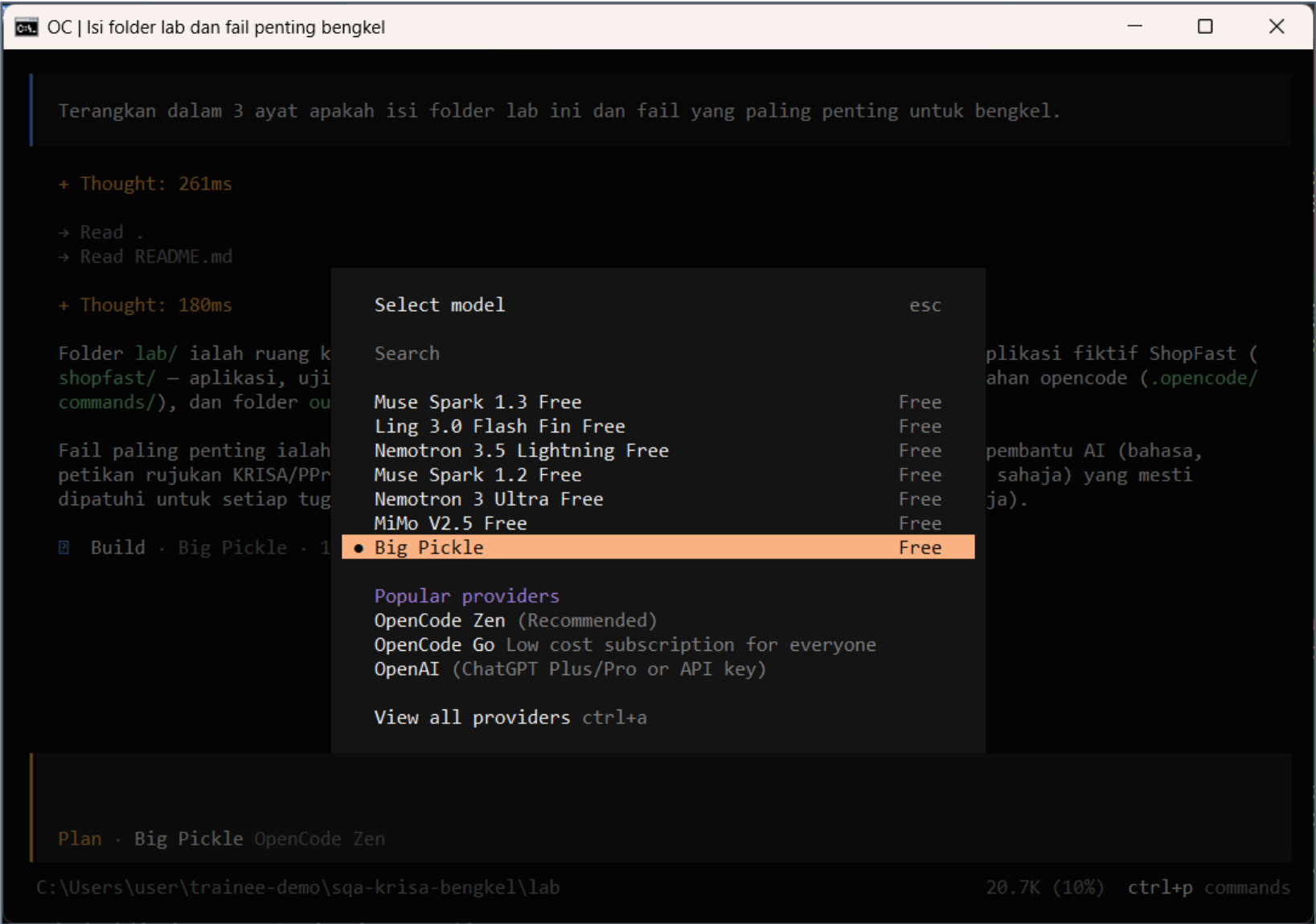


/models shows the free models. Keep Big Pickle unless the trainer says otherwise.

/models memaparkan model percuma. Kekalkan Big Pickle melainkan jurulatih mengarahkan lain.

TYPE IN OPENCODE

```
/models
```

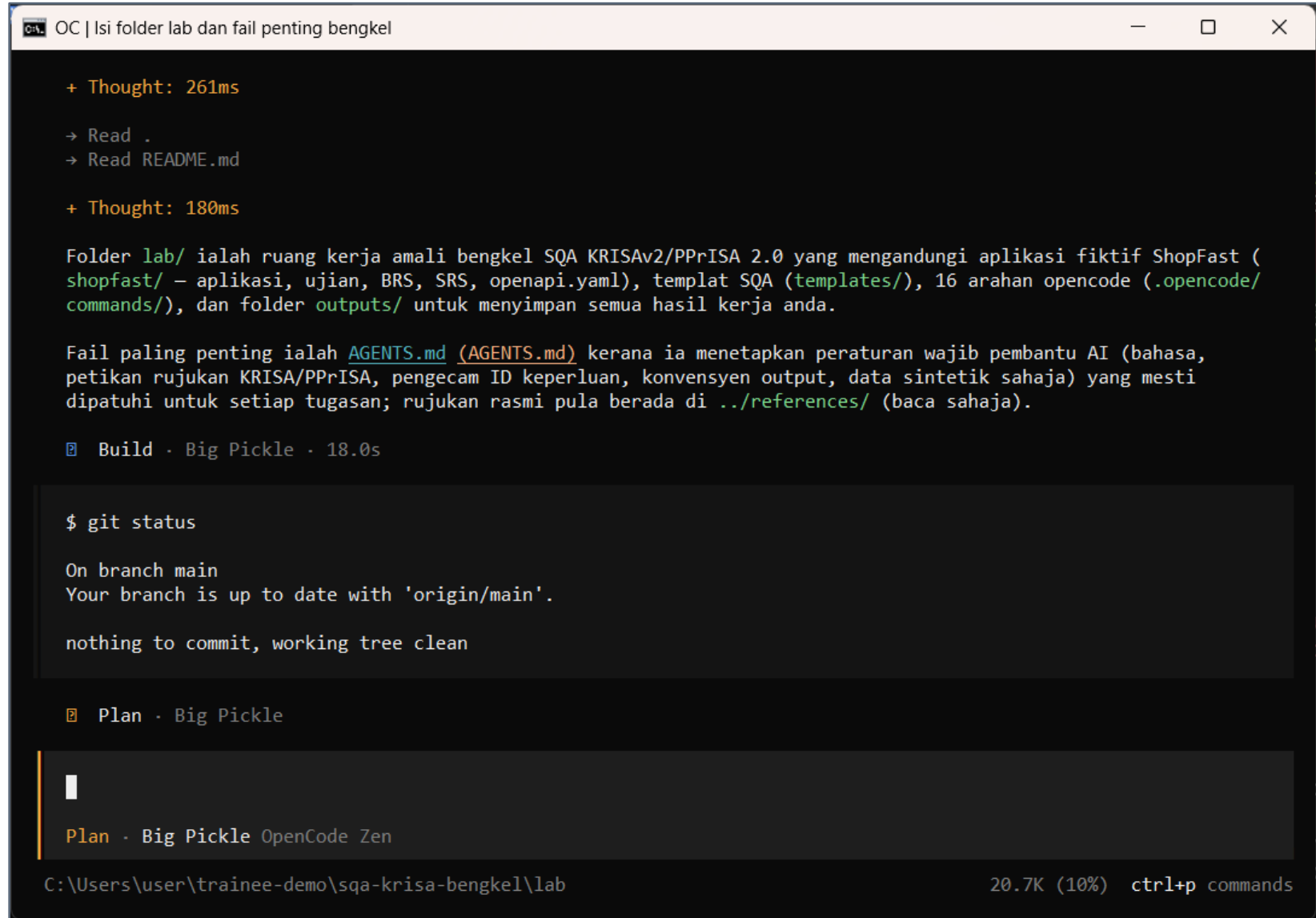


Start a line with ! to run a terminal command without leaving opencode.

Mulakan baris dengan ! untuk menjalankan arahan terminal tanpa keluar dari opencode.

TYPE IN OPENCODE

```
!git status
```



The screenshot shows the OpenCode IDE interface. The title bar reads "OC | Isi folder lab dan fail penting bengkel". The main editor area contains the following text:

```
+ Thought: 261ms
→ Read .
→ Read README.md
+ Thought: 180ms

Folder lab/ ialah ruang kerja amali bengkel SQA KRISAv2/PPrISA 2.0 yang mengandungi aplikasi fiktif ShopFast (shopfast/ – aplikasi, ujian, BRS, SRS, openapi.yaml), templat SQA (templates/), 16 arahan opencode (.opencode/commands/), dan folder outputs/ untuk menyimpan semua hasil kerja anda.

Fail paling penting ialah AGENTS.md (AGENTS.md) kerana ia menetapkan peraturan wajib pembantu AI (bahasa, petikan rujukan KRISA/PPrISA, pengecam ID keperluan, konvensyen output, data sintetik sahaja) yang mesti dipatuhi untuk setiap tugas; rujukan rasmi pula berada di ../references/ (baca sahaja).
```

Below the editor, there is a "Build" section showing "Build · Big Pickle · 18.0s". The terminal window displays the output of the command `$ git status`:

```
$ git status

On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
```

Below the terminal, there is a "Plan" section showing "Plan · Big Pickle". At the bottom of the IDE, the status bar shows the file path `C:\Users\user\trainee-demo\sqa-krisa-bengkel\lab`, the file size `20.7K (10%)`, and the command `ctrl+p commands`.

Human check before you accept the output / Semakan manusia

check-lab.ps1 shows all OK

check-lab.ps1 menunjukkan SEMUA OK

opencode --version prints 1.18.x

opencode --version memaparkan 1.18.x

The status line shows Big Pickle, OpenCode Zen

Baris status memaparkan Big Pickle, OpenCode Zen

The AI draft becomes evidence only after this check. Record who checked it and when (Module 7 audit trail).

If something goes wrong / Jika ada masalah

opencode is not recognised

Close and reopen PowerShell so PATH refreshes. Still failing: `powershell -ExecutionPolicy Bypass -File .\setup\install-opencode.ps1 -Portable`

Scripts are disabled on this system

`Set-ExecutionPolicy -Scope CurrentUser RemoteSigned`, or run `opencode.cmd` instead of `opencode`

The model is slow or stops

Wait up to 2 minutes. Press Esc to interrupt, then send the prompt again. Try another free model with `/models`

Permission required dialog

Read the preview. Choose Allow once for files in `outputs/`. Reject anything outside the lab folder

File or command not found

Check the status bar path ends with `\lab`. Commands and `@files` are relative to the lab folder

Answer uses Indonesian words or dashes

Reply: Ikut AGENTS.md peraturan 1 dan 14, tulis semula dalam Bahasa Melayu Malaysia

Network or proxy error

`$env:HTTPS_PROXY = "http://proxy:port"` and `$env:NO_PROXY = "localhost,127.0.0.1"` before starting opencode

Port 3000 already in use

`$env:PORT = 3001` before npm start, and use `http://localhost:3001`

Abbreviations / Singkatan (1 of 2)

Short forms used in this workshop, in plain words. / Singkatan dalam bengkel ini, dengan bahasa mudah.

Abbreviation / Singkatan	Stands for / Maksud penuh	In plain words / Maksud mudah
AI	Artificial Intelligence	Software that can answer questions and write text or code by learning patterns from large amounts of data. <i>Kecerdasan Buatan. Perisian yang boleh menjawab soalan dan menulis teks atau kod berdasarkan pola daripada data yang banyak.</i>
API	Application Programming Interface	A doorway that lets one program ask another program for data or an action, without using a screen. <i>Pintu yang membolehkan satu program meminta data atau tindakan daripada program lain tanpa melalui skrin.</i>
KRISA	Kejuruteraan Sistem Aplikasi Sektor Awam	The Malaysian public sector guide for engineering application systems. KRISAv2 is the 2026 beta edition. <i>Buku Panduan Kejuruteraan Sistem Aplikasi Sektor Awam. KRISAv2 ialah edisi beta 2026.</i>
LTS	Long-Term Support	A software version that gets fixes for a long time. The safest choice for a lab PC, as in Node.js LTS. <i>Versi perisian yang menerima pembetulan untuk tempoh panjang. Pilihan paling selamat untuk PC makmal, seperti Node.js LTS.</i>
MyIPO	Perbadanan Harta Intelek Malaysia	Intellectual Property Corporation of Malaysia, the host of this training venue. <i>Perbadanan Harta Intelek Malaysia, tuan rumah lokasi latihan ini.</i>

Abbreviations / Singkatan (2 of 2)

Short forms used in this workshop, in plain words. / Singkatan dalam bengkel ini, dengan bahasa mudah.

Abbreviation / Singkatan	Stands for / Maksud penuh	In plain words / Maksud mudah
PC	Personal Computer	The lab computer you are working on. <i>Komputer peribadi, iaitu komputer makmal yang anda gunakan.</i>
PPrISA	Panduan Pengurusan Projek ICT Sektor Awam	The Malaysian public sector guide for managing ICT projects. Codes such as PPrISA05 are its templates. <i>Panduan Pengurusan Projek ICT Sektor Awam. Kod seperti PPrISA05 ialah templatnya.</i>
SQA	Software Quality Assurance	All the planned work that gives confidence the software meets its quality requirements. <i>Jaminan Kualiti Perisian, iaitu semua kerja terancang yang memberi keyakinan bahawa perisian memenuhi keperluan kualiti.</i>
TLS	Transport Layer Security	The encryption that protects web downloads (the S in HTTPS). Old PowerShell sometimes needs it switched on. <i>Penyulitan yang melindungi muat turun web (huruf S dalam HTTPS). PowerShell lama kadangkala perlu mengaktifkannya.</i>
TUI	Text User Interface	A full-screen program that runs inside a terminal window, like opencode. <i>Antara muka pengguna berasaskan teks, iaitu program skrin penuh dalam tetingkap terminal seperti opencode.</i>

Links / Pautan

[Web guide with copyable prompts](#)

[Module hub \(slides, references, templates\)](#)

[Example outputs from the real run](#)

[Lab README and command list](#)

[AGENTS.md rules used by the assistant](#)

[opencode documentation](#)

[opencode TUI commands and keys](#)

[OpenCode Zen free models and data policy](#)